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딥러닝 영상 분석을 활용한
클라이밍 경쟁 서비스

Team #5: 못올라가요 (나종우, 장원범, 조준오)

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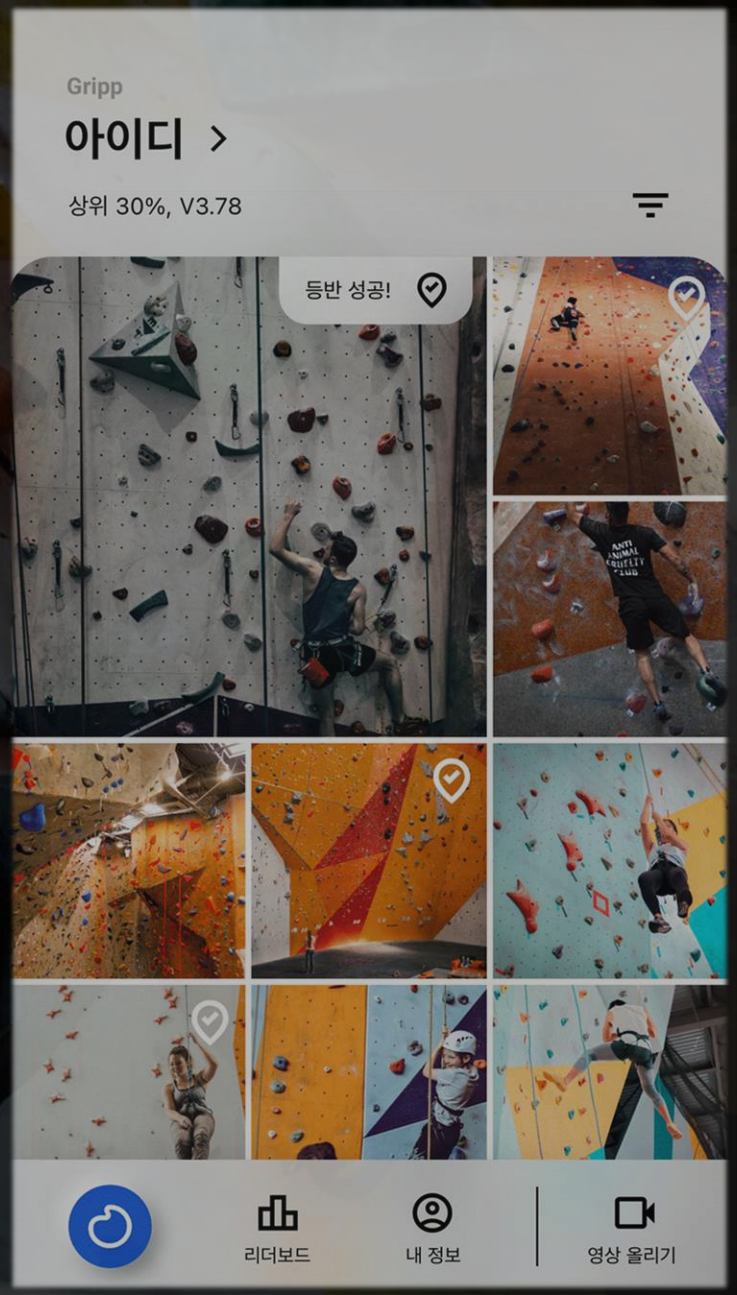

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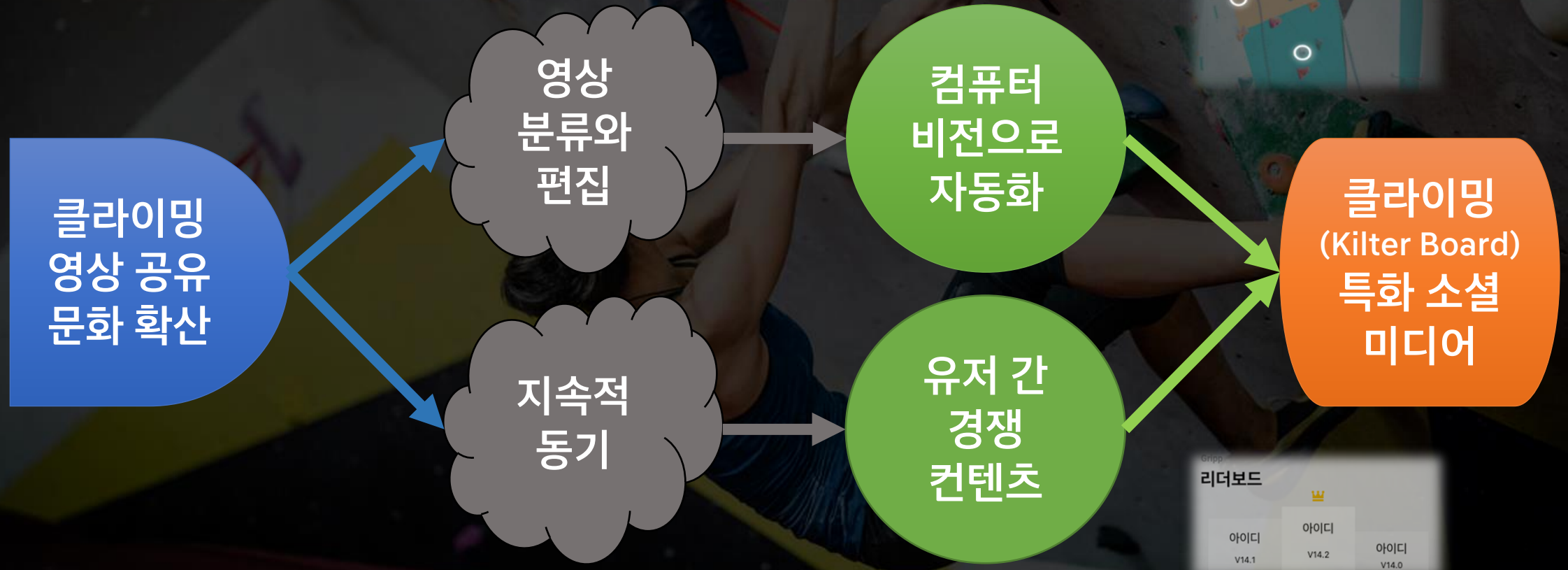
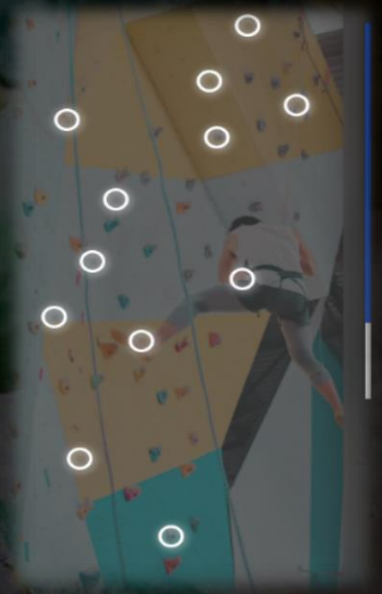
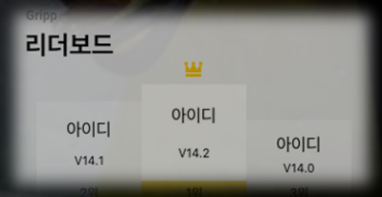
Gripp = Grip + People







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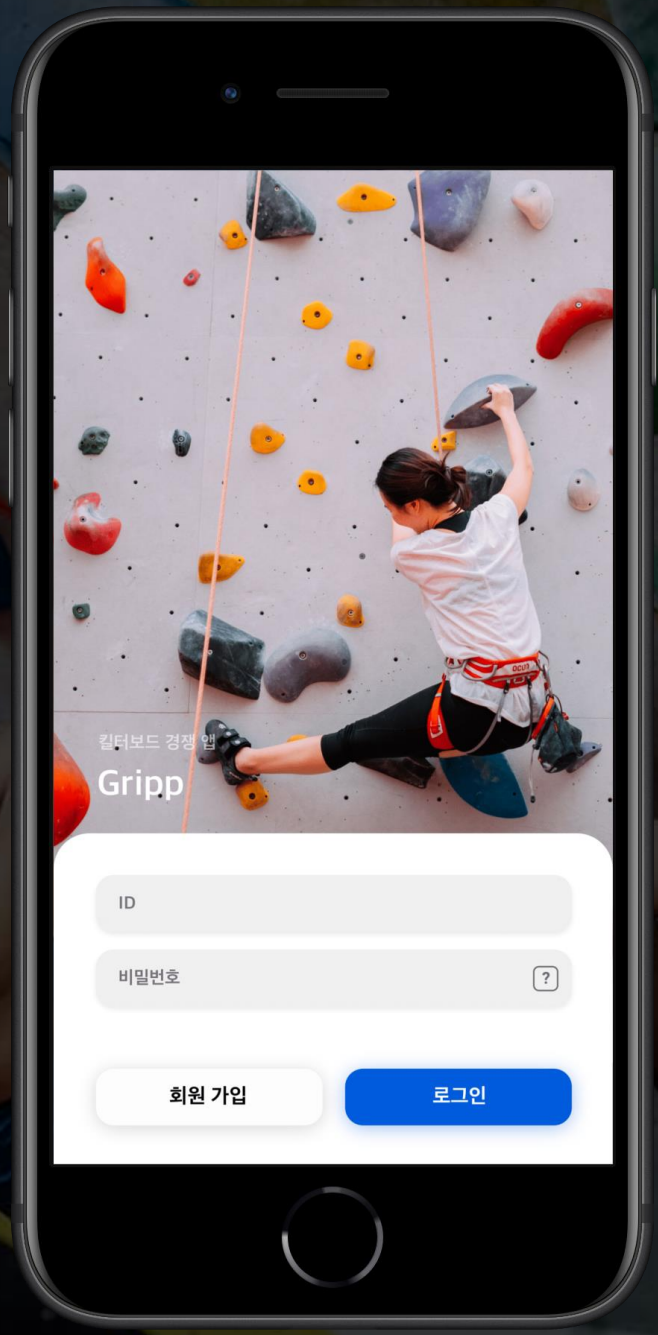
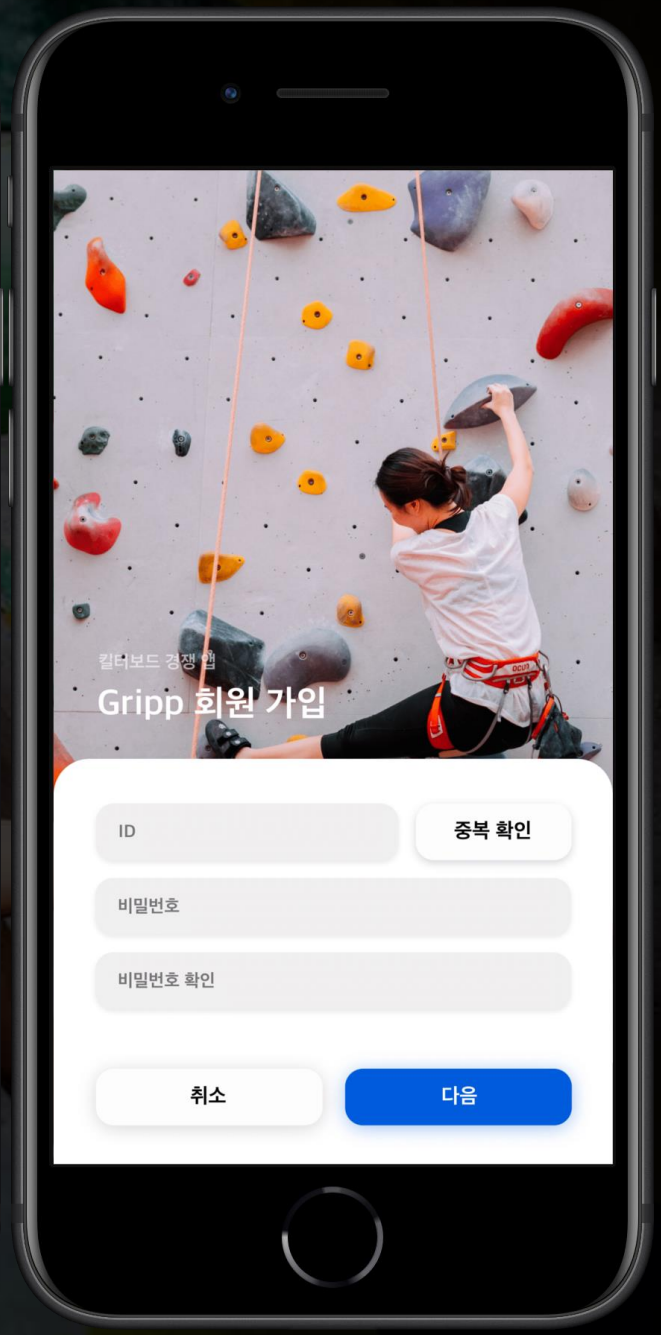
로그인

유저와 마주하는 첫 화면

- 계정 관련 API 개발 완료
- 모든 API에 인증, 인가 적용
- SwiftUI 구성 완료





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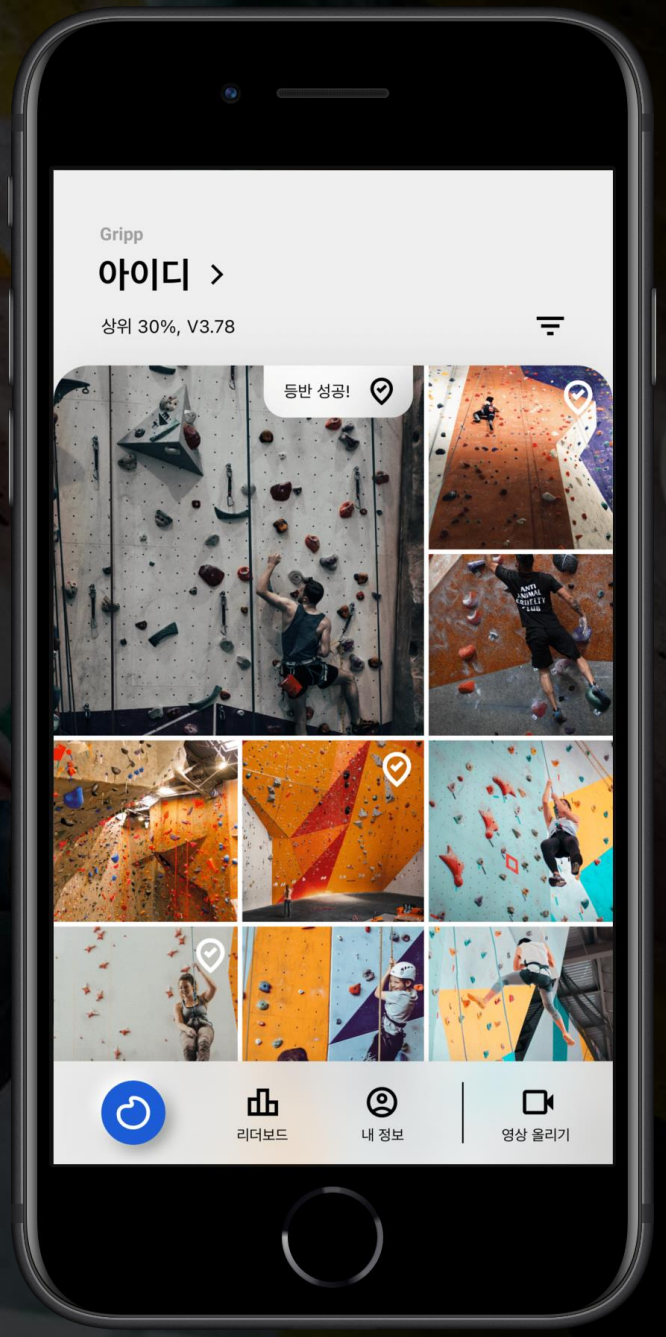
홈

게시물을 둘러보는 피드

- 게시물 검색 API 개발 중
  - 영상 썸네일
  - 영상 분석 중 / 등반 성공 / 등반 실패 상태
  - 난이도 필터
- SwiftUI 구성 완료



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게시물 보기

클라이밍(Kilter Board) 영상을 시청하는 화면

- SwiftUI 구성 완료

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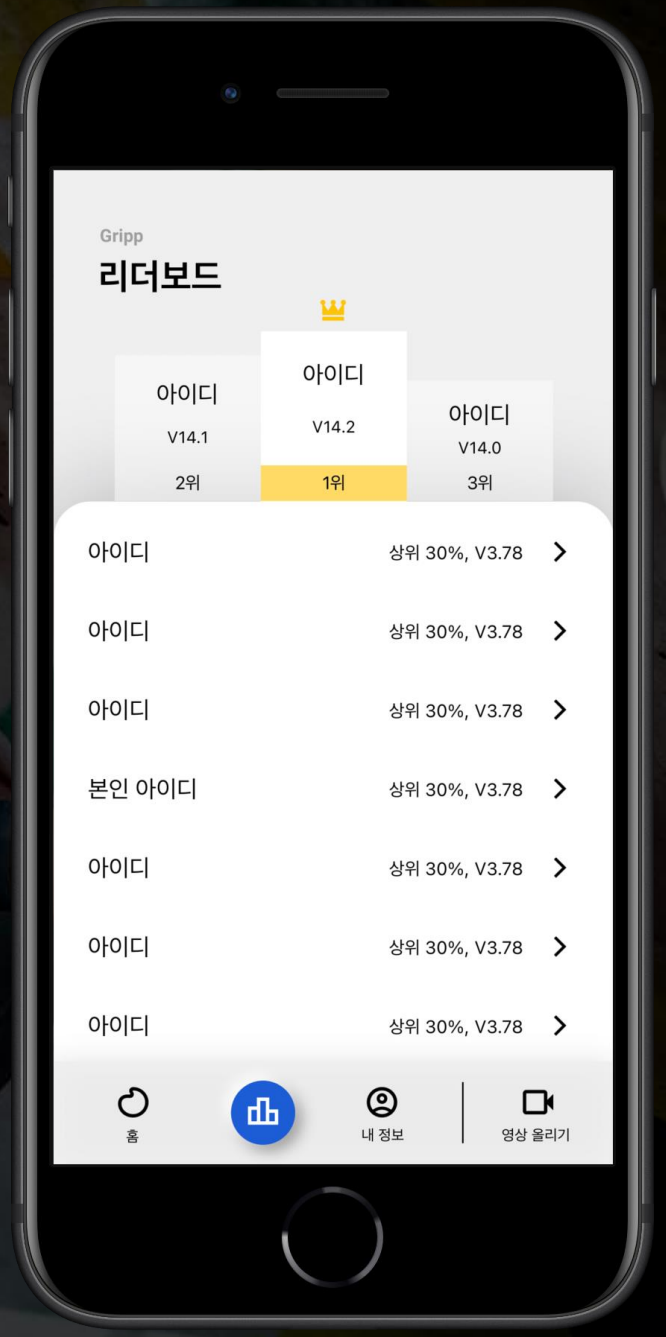
리더보드

유저들의 클라이밍 실력 경쟁

- 리더보드 API 개발 완료
  - 1) 상위 리더보드 2) 내 근처 리더보드
  - 점수 : 등반 성공 영상만 고난도 순으로 10개까지 난이도 가중치 합산 (0.00 ~ 19.00)
  - 티어 : 점수를 반올림한 값 (V0 ~ V19)
- SwiftUI 구성 완료



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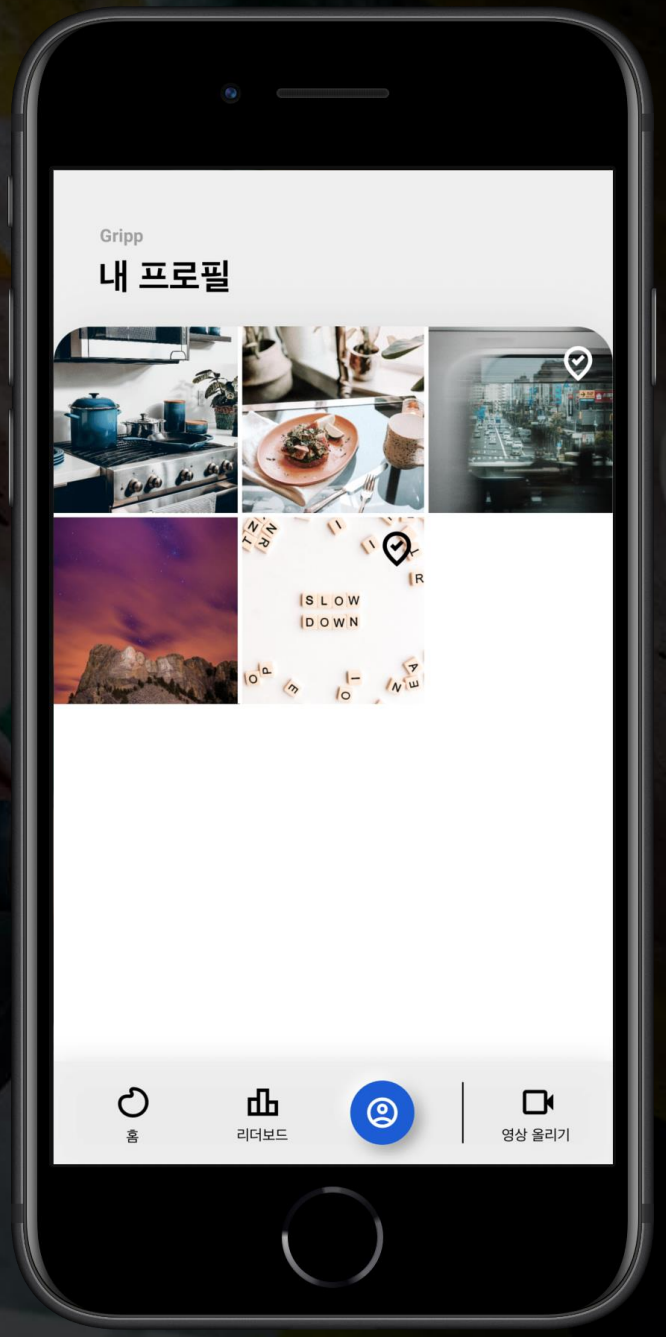
내 정보

나의 프로필과 게시물을 보는 화면

- 회원정보 API 개발 완료
  - 점수, 티어, 순위, 백분위, 게시물 수, 등반 성공 수
- 게시물 검색 API 개발 중
  - 유저 필터
- SwiftUI 구성 완료



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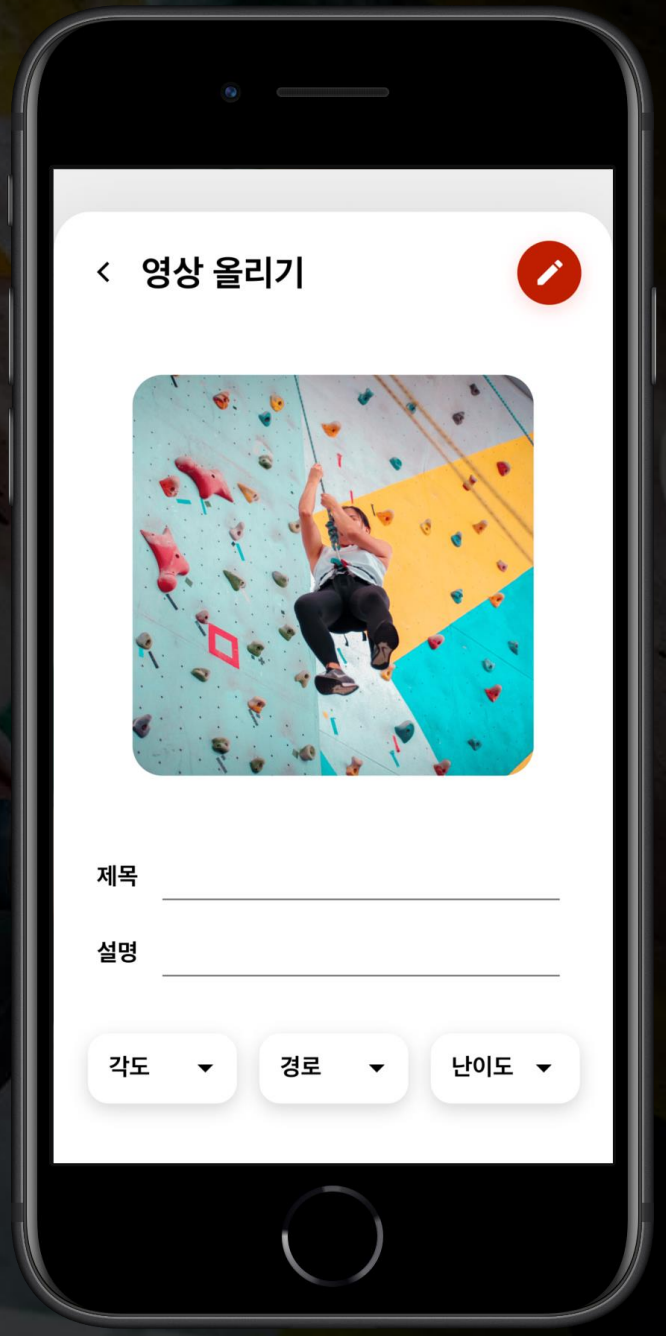
영상 올리기

휴대폰에서 찍은 클라이밍(Kilter Board) 영상 업로드

- <u>AI 영상 분류</u>
  - (등반 성공 or 등반 실패) 판정
- <u>AI 영상 편집</u>
  - (등반 시작 시점 ~ 종료 시점)을 인식해 CUT 편집



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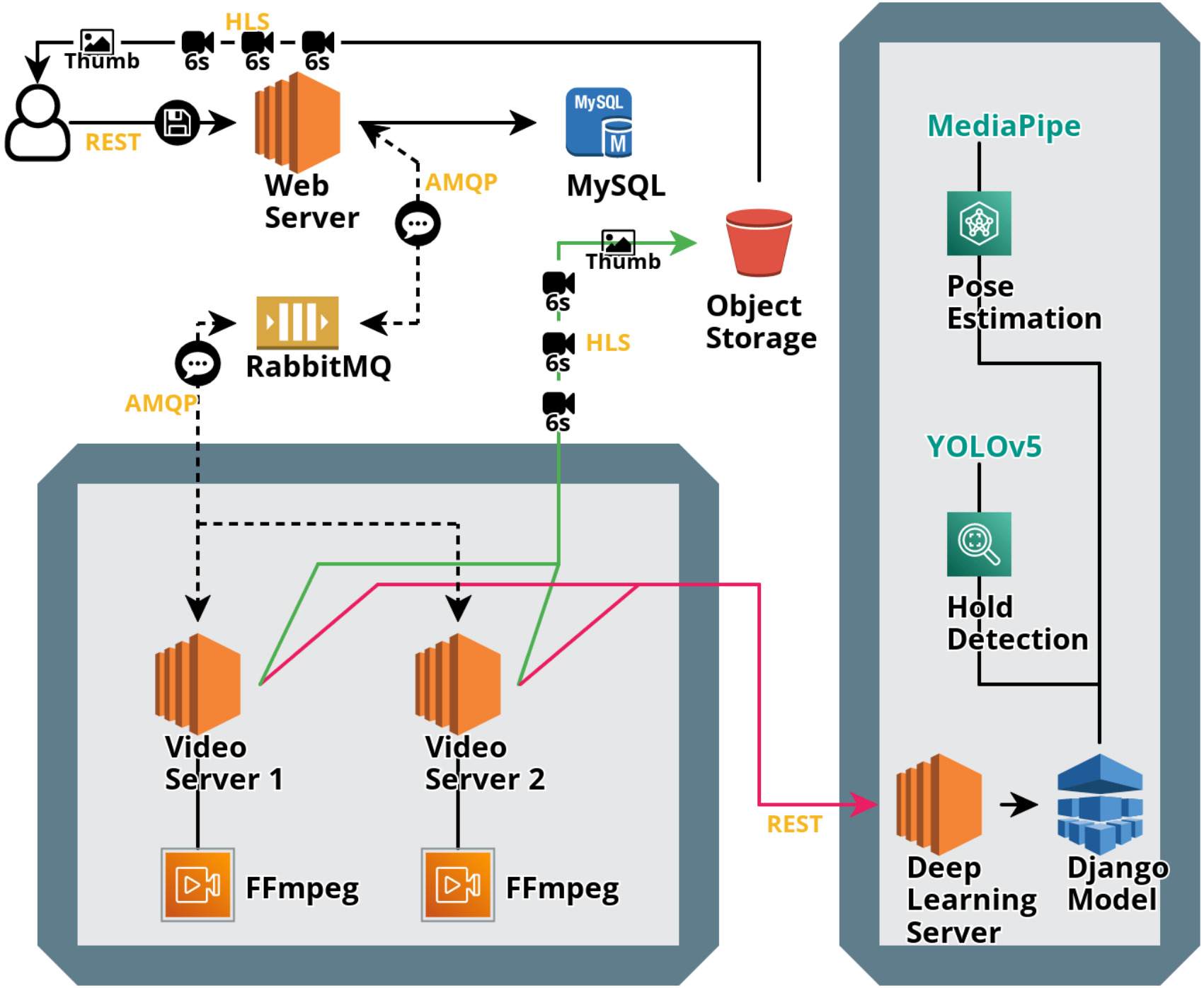
영상 올리기


Thumb
HLS
6s
6s
6s
REST
Web Server
AMQP
MySQL
MySQL
Thumb
6s
HLS
6s
6s
Object Storage
RabbitMQ
AMQP
Video Server 1
Video Server 2
FFmpeg
FFmpeg
REST
MediaPipe
Pose Estimation
YOLOv5
Hold Detection
Deep Learning Server
Django Model

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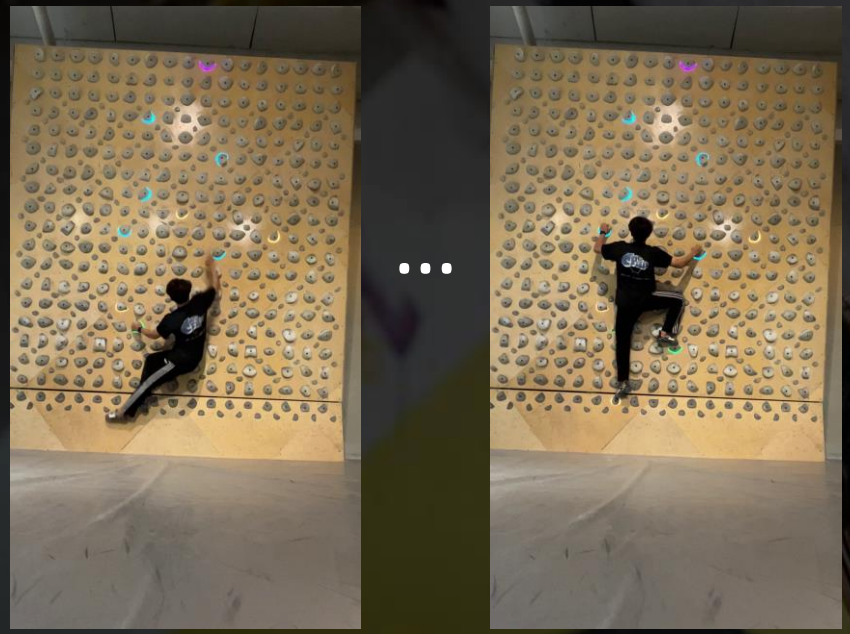
영상 분석

1) Hold Detection

Start Hold : 시작 지점
Hand Hold : 손과 발로 디딜 수 있음
Foot Hold : 발만 디딜 수 있음
Top Hold : 도착 지점



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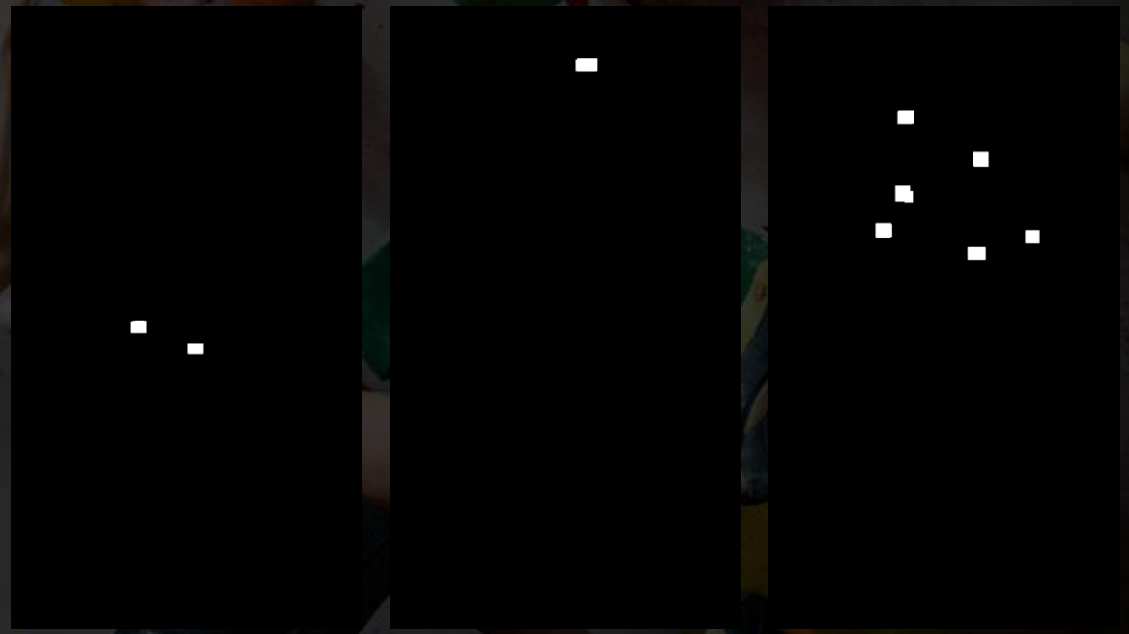

start hold

top hold

hand hold

비디오에서 이미지 5장 추출

5개 Mask Image 생성 후 bitwise OR

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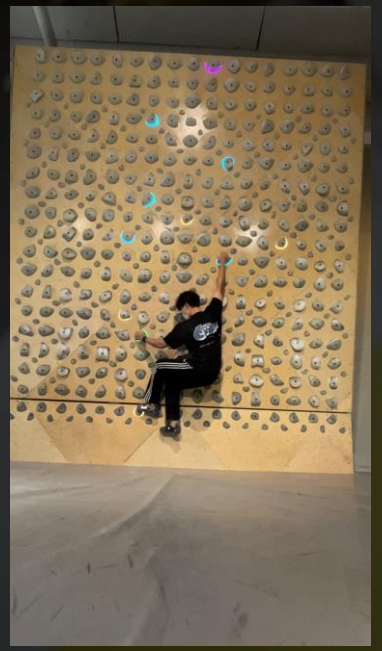
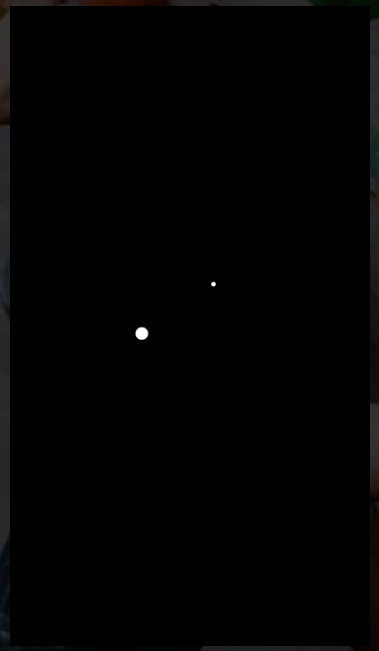
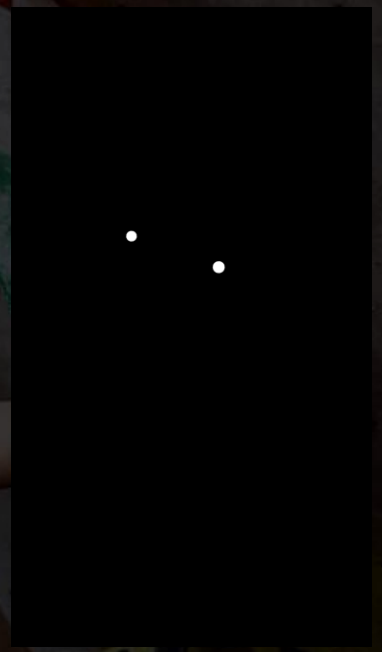
영상 분석

2) Pose Estimation (Hands)







0.5초마다 이미지 추출

손 Mask 생성

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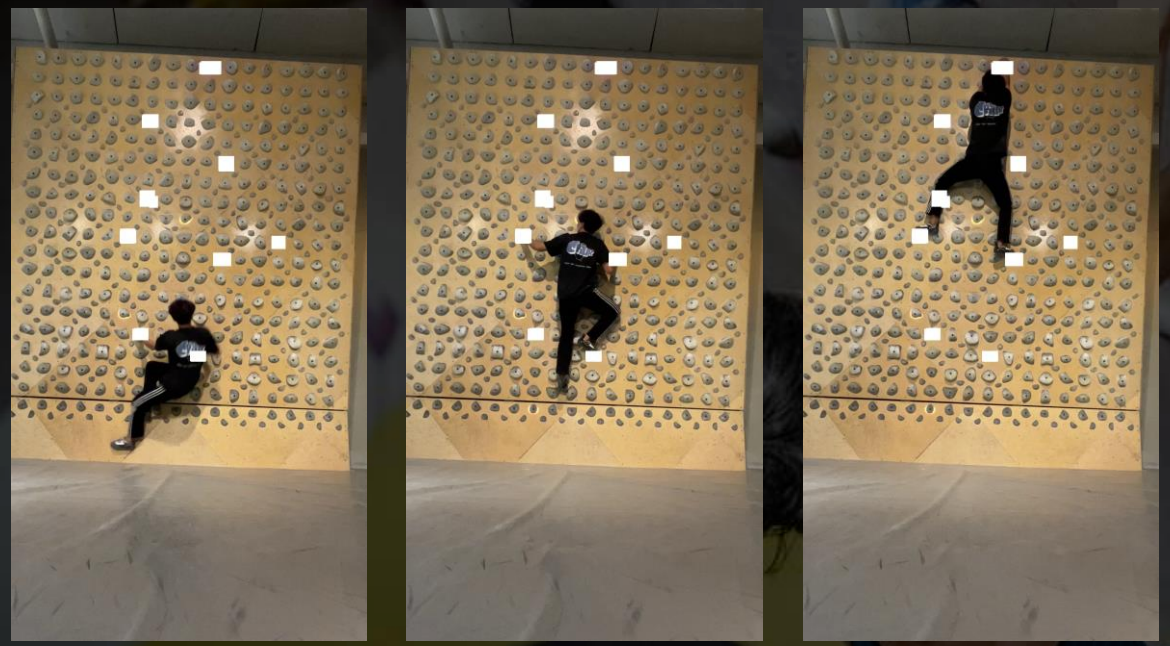
영상 분석

3) Final Algorithm



```
FPS = ?? (영상의 FPS)
frame_interval = FPS / 2 (0.5초마다 추출)
start_wait = end_wait = 1.5 * FPS / frame_interval (1.5초 대기)

success = false
start_frame = end_frame = 0

for each frame in video step frame_interval:
    if (hand mask & start hold mask):
        start_wait -= 1
        if (start_wait == 0):
            start_frame = frame

    if (hand mask & hand hold mask):
        end_frame = frame

    if (hand mask & top hold mask):
        end_wait -= 1
        if (end_wait == 0):
            success = true
            end_frame = frame

return (success, start_frame, end_frame)
```

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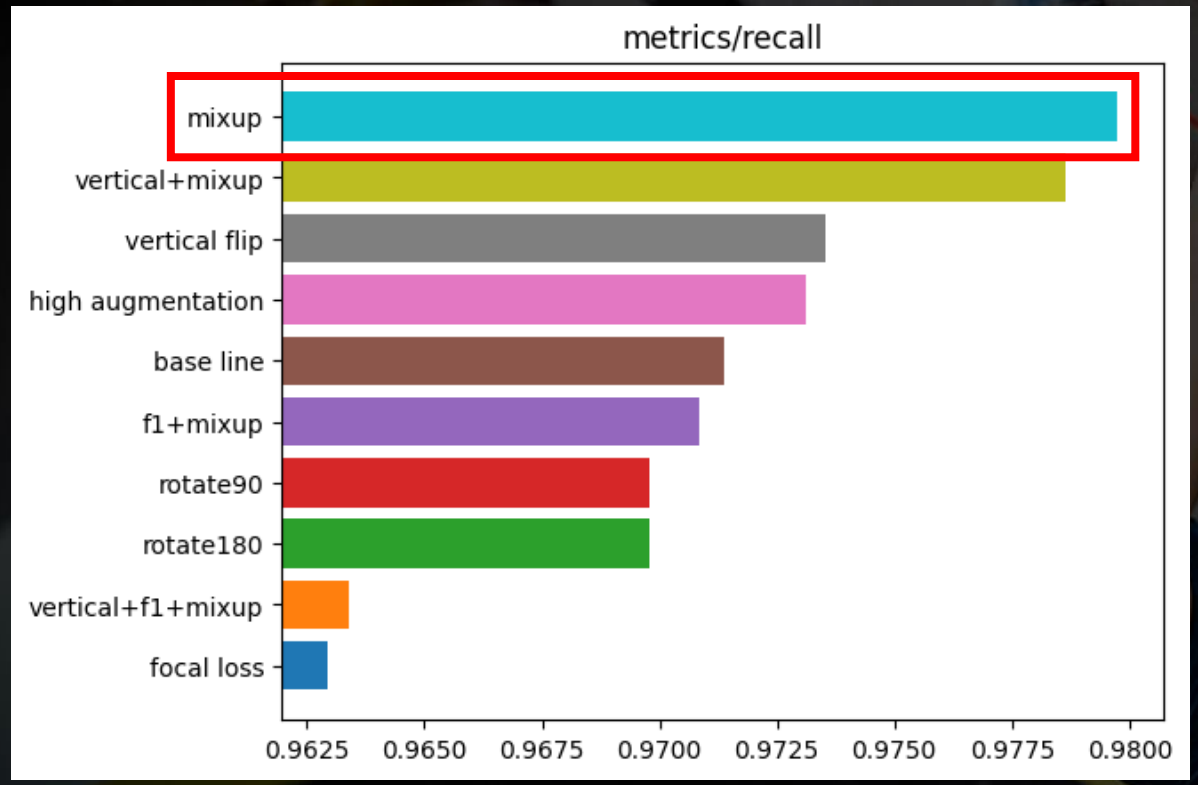
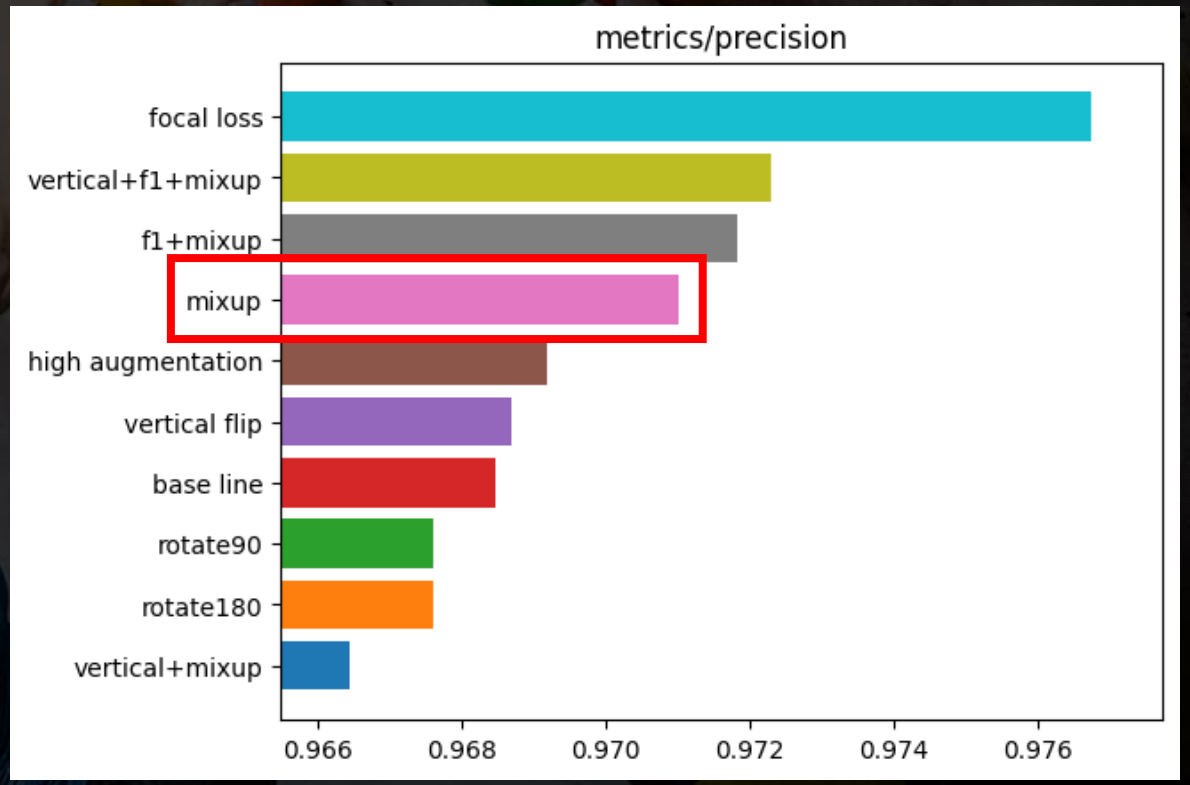
영상 분석

4) Augmentation





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영상 분석

4) Augmentation

Mosaic

Mosaic + MixUp

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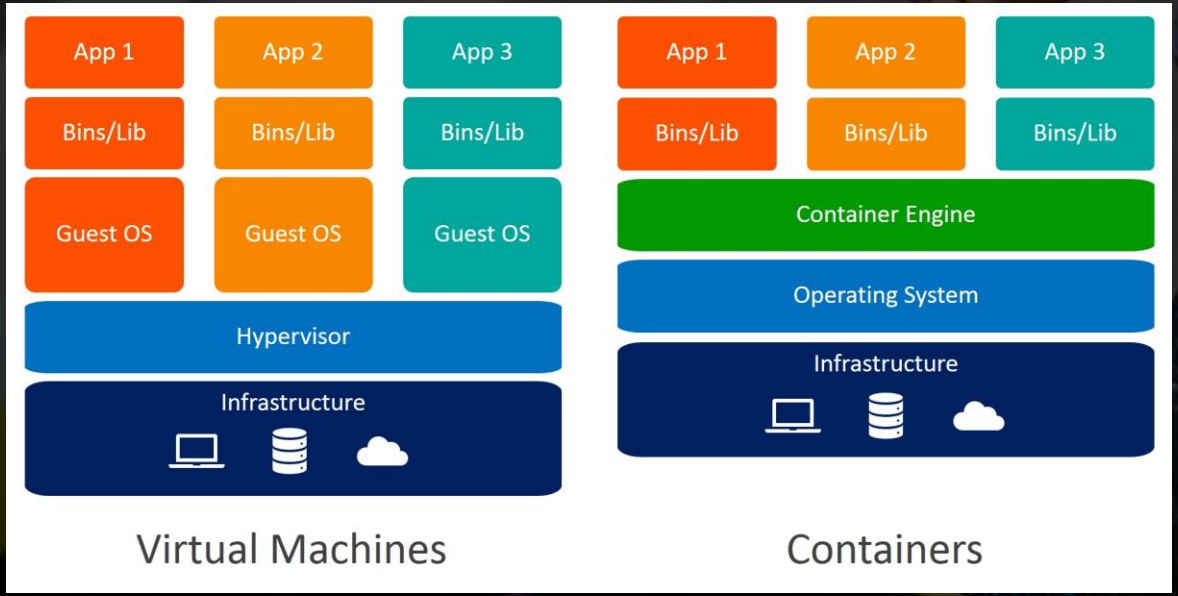
기타

- 비디오 서버, 딥러닝 서버 배포 시작 (Oracle Cloud)
  - Docker를 사용해 컨테이너로 관리



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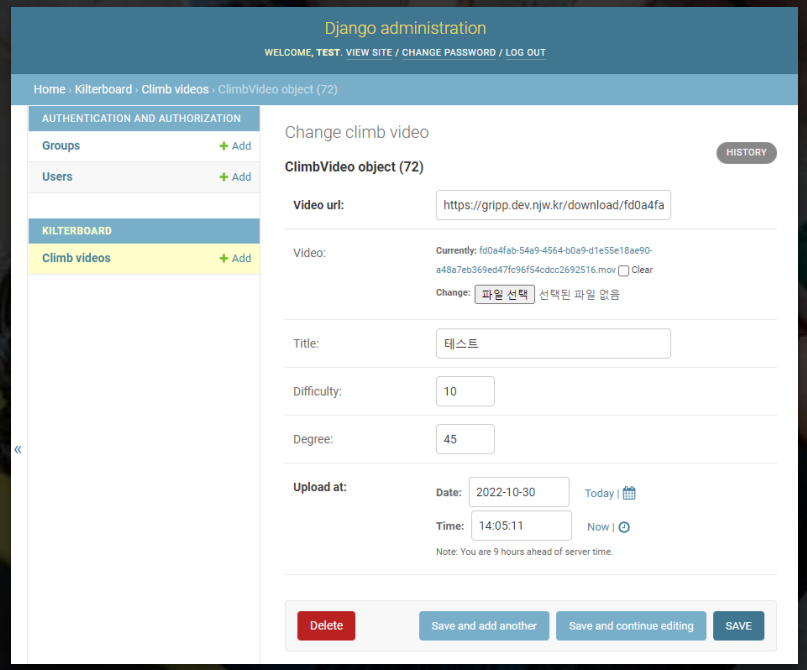
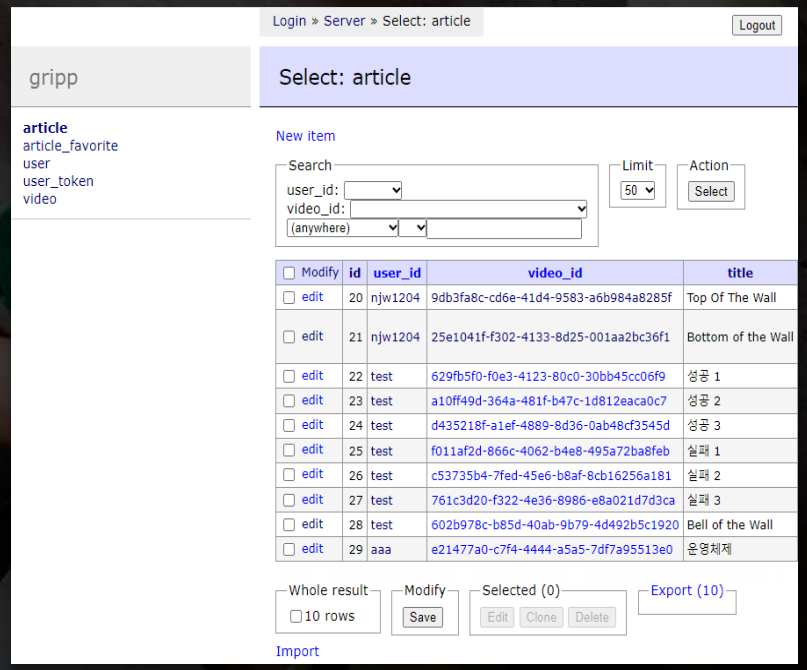
기타

- 내부 관리툴 3종 제작

모니터링 대시보드
(w/ Prometheus, Grafana)



딥러닝 서버 관리툴
(w/ Django Admin)



애플리케이션 관리툴
(w/ Adminer Editor)

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시연

- 영상 업로드 API
- iOS 앱 (SNAPSHOT 버전)

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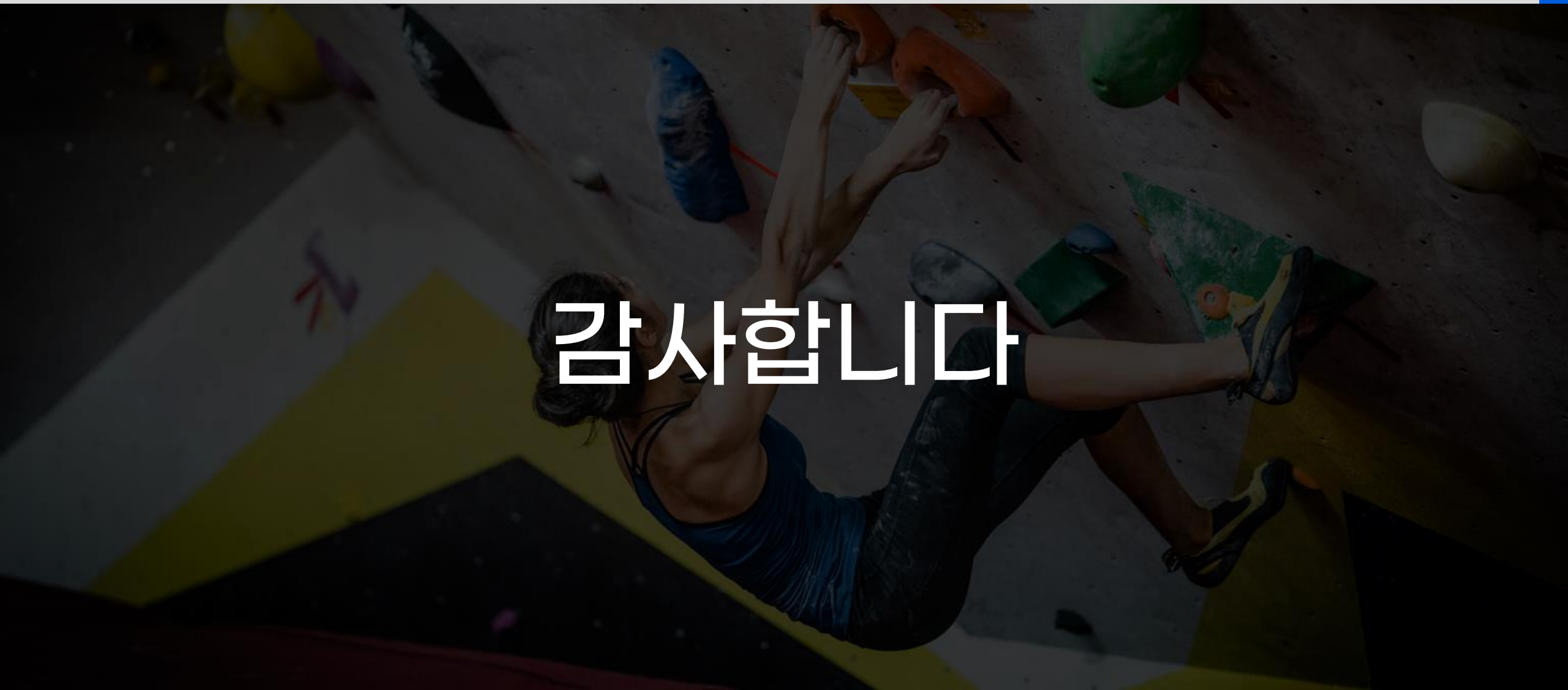

감사합니다